CLI-Based Expense Tracker

Project Report

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1. Introduction

Managing daily expenses manually can be inefficient and error-prone. Many individuals find it difficult to track where their money is being spent. This project aims to provide a simple and effective solution using a command-line interface to record and analyze daily expenses.

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2. Problem Statement

Keeping track of daily expenses without a structured system leads to poor financial awareness. There is a need for a lightweight and easy-to-use tool that allows users to log and review their expenses efficiently.

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3. Objective

The main objectives of this project are:

- To develop a command-line application for expense tracking
- To allow users to add and view expenses
- To provide category-wise summaries of spending
- To store data persistently using a file-based system

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4. Methodology

The project is implemented using Python and follows a modular programming approach. The application consists of multiple files handling different responsibilities:

- **main.py**: Controls the user interface and menu system

- **expense.py**: Contains logic for adding, viewing, and summarizing expenses
- **storage.py**: Handles reading and writing data to a JSON file

The data is stored in a JSON file (data.json), which ensures persistence between program runs.

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5. Features

- Add new expenses with amount and category
- View all recorded expenses
- Display category-wise expense summary
- Automatic date tracking
- Persistent storage using JSON

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6. Tools and Technologies Used

- Programming Language: Python
- Data Storage: JSON
- Platform: Command Line Interface (CLI)

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7. Implementation Details

The application provides a menu-driven interface where users can choose options to perform different actions. Based on user input, the program calls appropriate functions to process and display data. The use of JSON allows easy storage and retrieval of expense records.

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8. Results

The project successfully allows users to:

- Record daily expenses
- View transaction history
- Analyze spending patterns by category

The application runs efficiently in a terminal environment without requiring any graphical interface.

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9. Challenges Faced

- Handling file operations and ensuring data persistence
- Designing a user-friendly CLI interface
- Managing errors such as invalid inputs

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10. Learning Outcomes

- Learned how to build a command-line application
- Gained experience in file handling using JSON
- Understood modular programming in Python
- Improved problem-solving and debugging skills

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11. Conclusion

The CLI-Based Expense Tracker provides a simple yet effective solution for managing daily expenses. It demonstrates the practical application of Python programming concepts such as file handling, modular design, and user interaction.

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12. Future Enhancements

- Add monthly and yearly reports
- Implement data visualization (graphs)
- Add user authentication
- Convert into a web-based or mobile application

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13. References

- Python Official Documentation
- JSON Documentation